

Chapter 1 Test

September 6, 2020

Problem 1

A rational number is a number formed by dividing two integers $\frac{a}{b}$ where $b \neq 0$. The set of all rational numbers is denoted $\mathbb{Q}$. Explain why every integer is a rational number, but not all rational numbers are integers.

Problem 2

Each of the following statements can be recast in the if-then form. Please rewrite each of the following sentences in the form “If A, then B.”

1. The product of an odd integer and an even integer is even.
2. The square of an odd integer is odd.
3. The square of a prime number is not prime.
4. The sum of three consecutive integers is divisible by three.

Problem 3

Prove that the sum of an odd integer and an even integer is odd.

Problem 4

Suppose a , b , and c are integers. Prove that if $a|b$ and $a|c$, then $a|(b + c)$.

Problem 5

Disprove: If a , b , and c are positive integers with $a|(bc)$ then $a|b$ or $a|c$.

Problem 6

Prove: $(x \wedge y) \vee (x \wedge \neg y)$ is logically equivalent to x .

Hint: use a truth table.

Problem 7

Determine which of the following statements are true and which are false. You should base your reply on your common knowledge of mathematics; you do not need to prove your answers.

1. Every positive integer is either prime or composite.
2. Let x and y be integers. We have $x^2 = y^2$ if and only if $x = y$.
3. If x is an integer and $x > 2$ and x is prime, then x is odd.

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Problem 8

A *tautology* is a Boolean expression that evaluates to *true* for all possible values of its variables. For example, the expression $x \vee \neg x$ is *true* both when $x = \text{true}$ and when $x = \text{false}$. Thus $x \vee \neg x$ is a tautology.

Is the Boolean expression $(x \rightarrow y) \vee (x \rightarrow \neg y)$ a tautology? Justify your answer.

Extra Credit:

What can you say about the sum of consecutive perfect cubes, starting with 1. That is, what can you say about 1 , $1 + 3^3$, $1 + 3^3 + 5^3$, $1 + 3^3 + 5^3 + 7^3$, and so on.